

CODEALPHA

BIOINFORMATICS INTERNSHIP

TASK 2

MULTIPLE SEQUENCE ALIGNMENT OF HEMOGLOBIN BETA

Five beta-globin proteins | Clustal Omega alignment | Conserved motifs

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Abstract

A multiple sequence alignment was prepared for five reviewed beta-globin proteins from human, rhesus macaque, mouse, rat and mallard. Human HBB (P68871) was used as the reference. The alignment was evaluated using consensus conservation, pairwise identity and a neighbour-joining guide tree. The five sequences aligned across 147 positions, with 88 positions completely conserved and 116 positions showing at least 80% residue conservation. Human HBB shared 94.52% identity with rhesus macaque, 80.27% with mouse, 78.91% with rat and 69.39% with mallard. Conserved blocks surrounding WGKV, PWTQ, HGKKV, HCDKLHVPENF and the C-terminal HKYH pattern demonstrate preservation of the globin core and heme-associated functional environment.

1. Introduction

Multiple sequence alignment (MSA) places homologous residues from three or more sequences into common columns. Conserved columns can indicate structural constraint, catalytic or binding functions, and common evolutionary origin. Clustal Omega uses guide trees and profile-based methods to produce scalable protein alignments. In this task, five beta-globin proteins were aligned to identify conserved regions, compare sequence identities and interpret functionally important motifs.

2. Sequence Dataset

Accession	Entry	Organism	Gene	Length
P68871	HBB_HUMAN	Homo sapiens	HBB	147
P02026	HBB_MACMU	Macaca mulatta	HBB	146
P02088	HBB1_MOUSE	Mus musculus	Hbb-b1	147
P11517	HBB2_RAT	Rattus norvegicus	Hbb-b2	147
P02114	HBB_ANAPL	Anas platyrhynchos	HBB	147

All sequences are members of the hemoglobin beta family. The dataset includes closely related primate proteins, two rodent proteins and one avian protein, providing a useful range of evolutionary distances.

3. Methodology

The five FASTA sequences were combined in a single file with unique identifiers and aligned in protein mode using Clustal Omega. The alignment was exported in Clustal format. Pairwise identity was calculated from aligned non-gap positions. Column conservation was calculated as the frequency of the most common residue at each position. A neighbour-joining guide tree was constructed from alignment-based identity distances. The tree is used as a similarity summary and is not interpreted as a fully supported phylogenetic reconstruction.

Input: five reviewed beta-globin sequences | Alignment length: 147 positions |
Reference: human P68871

Clustal Omega Multiple Sequence Alignment of Five Beta-Globin Proteins

Alignment positions 1-50

Human	M	V	H	L	T	P	E	E	K	S	A	V	T	A	L	W	G	K	V	N	V	D	E	V	G	G	E	A	L	G	R	L	L	V	V	Y	P	W	T	Q	R	F	F	E	S	F	G	D	L	S
Rhesus macaque	-	V	H	L	T	P	E	E	K	N	A	V	T	T	L	W	G	K	V	N	V	D	E	V	G	G	E	A	L	G	R	L	L	L	V	Y	P	W	T	Q	R	F	F	E	S	F	G	D	L	S
Mouse	M	V	H	L	T	D	A	E	K	A	A	V	S	C	L	W	G	K	V	N	S	D	E	V	G	G	E	A	L	G	R	L	L	V	V	Y	P	W	T	Q	R	Y	F	D	S	F	G	D	L	S
Rat	M	V	H	L	T	D	A	E	K	A	T	V	S	G	L	W	G	K	V	N	A	D	N	V	G	A	E	A	L	G	R	L	L	V	V	Y	P	W	T	Q	R	Y	F	S	K	F	G	D	L	S
Mallard	M	V	H	W	T	A	E	E	K	Q	L	I	T	G	L	W	G	K	V	N	V	A	D	C	G	A	E	A	L	A	R	L	L	I	V	Y	P	W	T	Q	R	F	F	A	S	F	G	N	L	S

Alignment positions 51-100

Human	T	P	D	A	V	M	G	N	P	K	V	K	A	H	G	K	K	V	L	G	A	F	S	D	G	L	A	H	L	D	N	L	K	G	T	F	A	T	L	S	E	L	H	C	D	K	L	H	V	D
Rhesus macaque	S	P	D	A	V	M	G	N	P	K	V	K	A	H	G	K	K	V	L	G	A	F	S	D	G	L	N	H	L	D	N	L	K	G	T	F	A	Q	L	S	E	L	H	C	D	K	L	H	V	D
Mouse	S	A	S	A	I	M	G	N	A	K	V	K	A	H	G	K	K	V	I	T	A	F	N	D	G	L	N	H	L	D	S	L	K	G	T	F	A	S	L	S	E	L	H	C	D	K	L	H	V	D
Rat	S	A	S	A	I	M	G	N	P	Q	V	K	A	H	G	K	K	V	I	N	A	F	N	D	G	L	K	H	L	D	N	L	K	G	T	F	A	H	L	S	E	L	H	C	D	K	L	H	V	D
Mallard	S	P	T	A	I	L	G	N	P	M	V	R	A	H	G	K	K	V	L	T	S	F	G	D	A	V	K	N	L	D	N	I	K	N	T	F	A	Q	L	S	E	L	H	C	D	K	L	H	V	D

Alignment positions 101-147

Human	P	E	N	F	R	L	L	G	N	V	L	V	C	V	L	A	H	H	F	G	K	E	F	T	P	P	V	Q	A	A	Y	Q	K	V	V	A	G	V	A	N	A	L	A	H	K	Y	H
Rhesus macaque	P	E	N	F	K	L	L	G	N	V	L	V	C	V	L	A	H	H	F	G	K	E	F	T	P	Q	V	Q	A	A	Y	Q	K	V	V	A	G	V	A	N	A	L	A	H	K	Y	H
Mouse	P	E	N	F	R	L	L	G	N	M	I	V	I	V	L	G	H	H	L	G	K	D	F	T	P	A	A	Q	A	A	F	Q	K	V	V	A	G	V	A	T	A	L	A	H	K	Y	H
Rat	P	E	N	F	R	L	L	G	N	M	I	V	I	V	L	G	H	H	L	G	K	E	F	T	P	C	A	Q	A	A	F	Q	K	V	V	A	G	V	A	S	A	L	A	H	K	Y	H
Mallard	P	E	N	F	R	L	L	G	D ₁₁₀	L	I	I	V	L	A	A	H	F ₁₂₀	K	D	F	T	P	E	C	Q	A	A ₁₅₀	W	Q	K	L	V	R	V	V	A ₁₈₀	H	A	L	A	R	K	Y	H		

Figure 1. Multiple sequence alignment. Green columns are fully conserved; blue columns show at least 80% conservation.

4. Alignment Results

The alignment contains 88 completely conserved columns and 116 columns with conservation of 80% or greater. Only one leading gap is required, reflecting the absence of the initiating methionine in the rhesus macaque record. Most variation occurs through amino-acid substitutions rather than large insertions or deletions, which is typical of homologous globin chains that retain the same overall fold.

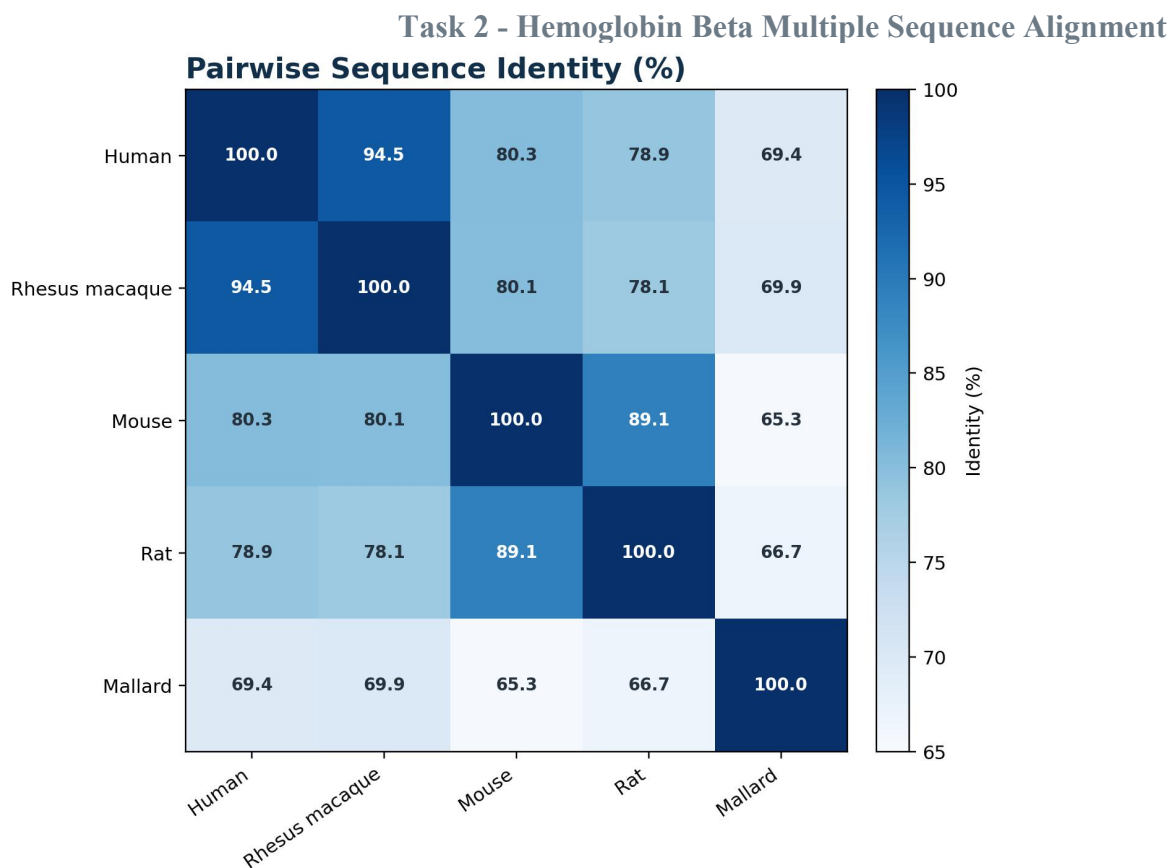
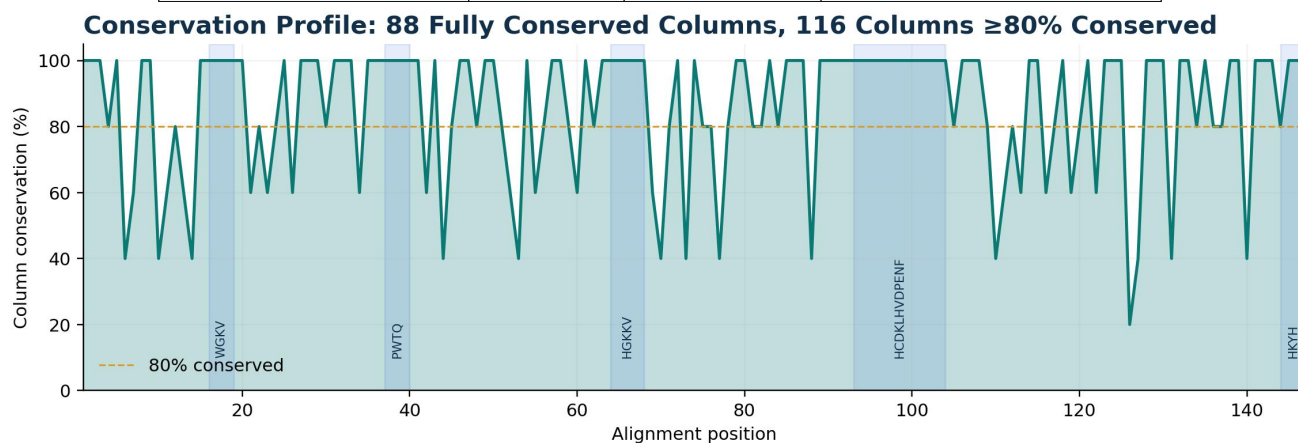


Figure 2. Pairwise percent identity calculated from aligned non-gap positions.

Comparison with human HBB	Identity	Alignment span	Interpretation
Rhesus macaque	94.52%	147 aa	Closest non-self homolog
Mouse	80.27%	147 aa	Mammalian homolog
Rat	78.91%	147 aa	Mammalian homolog
Mallard	69.39%	147 aa	Most distant homolog



Task 2 - Hemoglobin Beta Multiple Sequence Alignment

Figure 3. Per-column conservation profile with major conserved beta-globin sequence blocks marked.

5. Conserved Regions and Motifs

Motif/region	Human position	Alignment observation	Functional interpretation
WGKV	16-19	Identical in all five sequences	Conserved N-terminal globin region.
PWTQ	37-40	Retained in all five sequences	Hydrophobic/aromatic block contributing to the globin core.
HGKKV	64-68	Completely conserved	Contains the distal histidine region involved in ligand control.
HCDKLHVDPENF	93-104	Long highly conserved block	Contains the proximal histidine that coordinates heme iron.
HKYH	144-147	C-terminal pattern retained	Supports conservation of the terminal beta-globin structure.

The preservation of histidine-containing regions is particularly important. The distal histidine helps regulate access and stabilization of bound oxygen, whereas the proximal histidine forms the direct coordination bond with the heme iron. Conservation of their surrounding blocks supports preservation of beta-globin function across all five organisms.

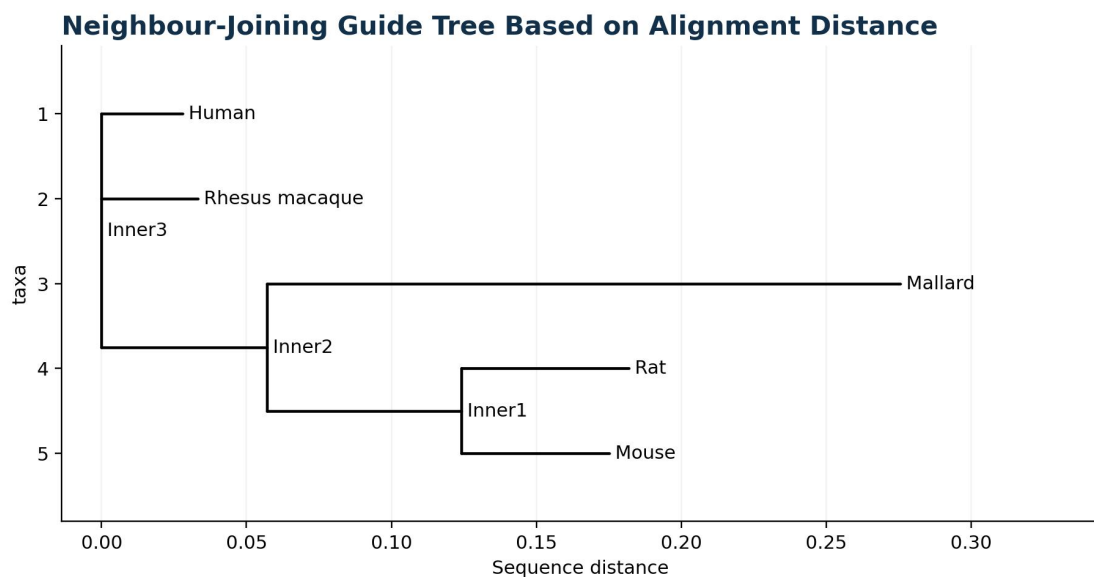


Figure 4. Alignment-based guide tree showing the closest clustering of human with rhesus macaque and mouse with rat.

6. Discussion

The MSA shows that beta-globin sequences remain alignable over their full length despite substantial evolutionary distance. Human and rhesus macaque form the closest pair, while mouse and rat form a second mammalian group. Mallard is the most distant sequence but still retains the major conserved motifs and high overall similarity. This pattern demonstrates that residues essential for the globin fold and heme interaction are under stronger selective constraint than surface positions that tolerate conservative substitutions.

The guide tree is consistent with the pairwise identity matrix, but it should not be treated as a definitive species phylogeny. A formal evolutionary analysis would require more sequences, explicit substitution models and statistical branch support. For this task, the tree appropriately summarizes similarity relationships within the selected protein family.

7. Conclusion

Clustal Omega MSA of five hemoglobin beta proteins identified extensive whole-sequence conservation and several functionally important motifs. Human HBB was most similar to rhesus macaque and progressively less similar to rodent and avian homologs. Nevertheless, the WGKV, PWTQ, HGKKV, HCDKLHVDPENF and C-terminal HKYH regions remained strongly conserved. The analysis demonstrates how MSA can reveal conserved residues, compare evolutionary relatedness and connect protein sequence patterns with structural function.

References

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